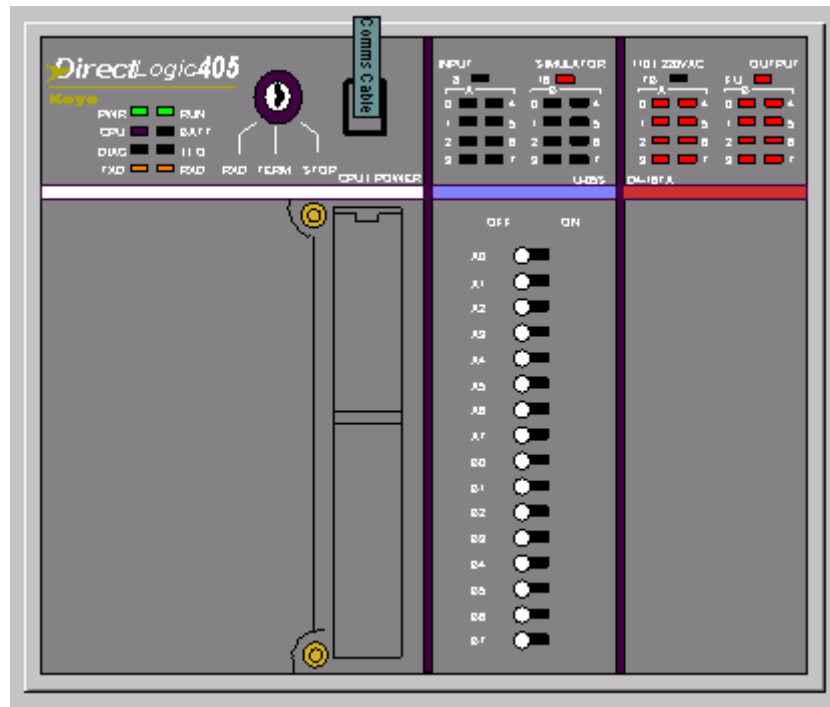


Koyo Direct Logic KDL405KS Protocol driver

Overview

Adroit supports communications with the Koyo Direct Logic DL405 PLC. Via the K-Sequence Protocol.



Device Configuration

KDL405KS Protocol Driver : BOILER1

Koyo Direct Logic 405 Details

CPU Model: DL430

Station ID: 1

Port Details

Comm Port: COM1

Baud rate: 9600

Parity: Odd

Stop Bits: One

Retries: 3

Timeout (ms): 1000

OK

Cancel

Help

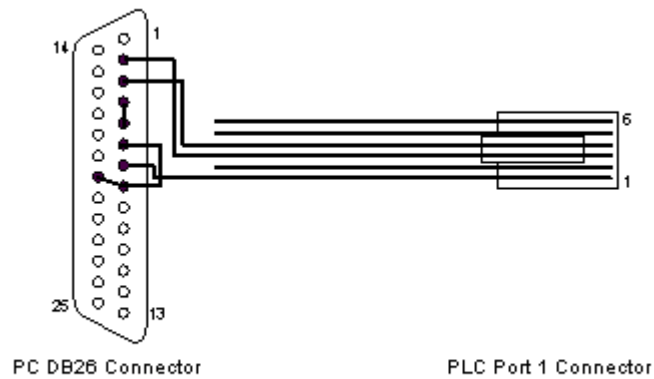
DL405 Details

CPU Model: The CPU Model currently installed in the base. 430, 440 or 450.
PLC Station ID: The ID number of the PLC on the PLC network.

Port Details

Port: The available serial ports on your machine
Baud Rate: The speed of data transmission for the PLC
 Up to 38400.
Parity: Check bit. None/Odd/Even. Default Odd.
Stop Bits: UART signal bits. One or Two. Default One.
Retries: The number of times that Adroit will try to gain communication with a PLC before failing a transaction.
Timeout: The time that Adroit will wait for a valid response from a PLC before failing a transaction (Read or Write).

Wiring



The diagram above shows the wiring for the comms cable between a PC serial port and the 450 CPU port 2.

For information on the use of other ports on the CPU's please refer to the DL405 User Manual 3rd Edition.

Supported addresses 430 CPU

Prefix	Start	End	Global Start	Global End	Type	Access	Description
TA	0	177	V0	177	WORD	Read Write	Timer Accumulator
CTA	0	177	V1000	1177	WORD	Read Write	Counter Accumulator
			V1400	7377	WORD	Read Write	Variable
			V7400	7777	WORD	Read Write	System Status
X	0	477	V40400	40423	BIT	Read Only	Input
Y	0	477	V40500	40523	BIT	Read Write	Output
C	0	737	V40600	40635	BIT	Read Write	Variable
S	0	577	V41000	41027	BIT	Read Write	Stage Status
T	0	177	V41100	41107	BIT	Read Write	Timer Status
CT	0	177	V41140	41147	BIT	Read Write	Counter Status
SP	0	137	V41200	41205	BIT	Read Only	System Status
SP	320	617	V41215	41230	BIT	Read Only	System Status

Supported addresses 440 CPU

Prefix	Start	End	Global Start	Global End	Type	Access	Description
TA	0	377	V0	377	WORD	Read Write	Timer Accumulator
CTA	0	177	V1000	177	WORD	Read Write	Counter Accumulator
			V1400	7377	WORD	Read Write	Variable
			V10000	17777	WORD	Read Write	Variable
			V700	737	WORD	Read Write	System Status
			V7400	7777	WORD	Read Write	System Status
X	0	477	V40400	40423	BIT	Read Only	Input
Y	0	477	V40500	40523	BIT	Read Write	Output
C	0	1777	V40600	40677	BIT	Read Write	Variable
S	0	1777	V41000	41077	BIT	Read Write	Stage Status
T	0	377	V41100	41117	BIT	Read Write	Timer Status
CT	0	177	V41140	41147	BIT	Read Write	Counter Status
SP	0	137	V41200	41204	BIT	Read Only	System Status
SP	320	717	V41215	41434	BIT	Read Only	System Status

Supported addresses 450 CPU

Prefix	Start	End	Global Start	Global End	Type	Access	Description
TA	0	377	V0	377	WORD	Read Write	Timer Accumulator
CTA	0	377	V1000	1377	WORD	Read Write	Counter Accumulator
			V1400	7377	WORD	Read Write	Variable
			V10000	37777	WORD	Read Write	Variable
			V700	777	WORD	Read Write	System Status
			V7400	7777	WORD	Read Write	System Status
X	0	1777	V40400	40477	BIT	Read Only	Input
Y	0	1777	V40500	40577	BIT	Read Write	Output
C	0	3777	V40600	40777	BIT	Read Write	Variable
S	0	1777	V41000	41077	BIT	Read Write	Stage Status
T	0	377	V41100	41117	BIT	Read Write	Timer Status
CT	0	377	V41140	41547	BIT	Read Write	Counter Status
SP	0	717	V41200	41237	BIT	Read Only	System Status

Adroit Data Types

Supported Adroit Data types are as follows:

PLC Data Type	Adroit Data Type		
	Bool	Integer	Real
WORD Read		√	√
WORD Write		√	√
BIT Read	√	√	√
BIT Write	√	√	√

Troubleshooting

NB All addresses with the Koyo DL405 PLC are expressed in Octal.

When reading or writing data to or from the DL405 the driver is actually always using 16 bit words. The driver, depending on the prefix, will work out the type of data and get or put only the correct information. For example when a digital input “X” prefix is used it can be mapped to any Adroit data type. The global equivalent for X0 is V40400. This is a word and X0 to X20 (The next 16 bits) are stored here. This can be read into any Adroit data type except BOOLEAN or any of its derivatives. The most efficient way of reading or writing digitals is to use a multistate agent and read the global equivalent address.

During the installation of the driver the following files will be copied:

KDL405KS.DLL	The driver itself.
KDL405KS.PDF	The driver help file.
KDL405KS.WGP	An Adroit database file.
KDL405KS.MIM	An adroit mimic file.
KDL405KS.REG	Registry file

Once the driver has been installed and setup, run the REG file, you may then set the Adroit database file to the KDL205KS.WGP and start the agent server.

When the agent server is ready, start the user interface and open the file KDL405KS.MIM. You will then see a representation of a DL405 base unit with a 450 CPU, an input simulation module and an output module.

The database will work with the 430 and 440 as well.

If all is installed correctly and the driver has successfully started, you will be able to see the inputs change when the simulator module switches are changed and also will be able to change output states when the output lights are highlighted and clicked.

This is the first thing to run when first setting up the unit. It can also be used as a training aid to see how the unit works.

Revision Information

Date	Changes
11-Jun-02	Adroit Technologies takes over driver from Alex Griffiths.
14-Aug-03	PDF format document distribution from now on